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Spatial cognition and emotion simulation in urban interior environments using agent-based modelling

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ABSTRACT

The proliferation of vertical urbanization has generated complex urban interior environments (UIEs), necessitating a deeper understanding of their impact on human well-being. This study investigates UIEs through neuroscience to develop a novel methodology for assessing volumetric spatial quality. This paper proposes the Cognitive-Emotional Volumetric Agent-Based Model (CEVABM), which integrates a systematic review of cognitive-emotional behavioural mechanisms to simulate spatial usage. By extrapolating emotional responses to UIE visual characteristics – including depth, colour and object taxonomies – into behavioural tendencies, CEVABM informs human flow simulations to evaluate perceived spatial quality. Results demonstrate quantifiable relationships between spatial features, emotional-cognitive responses and movement patterns, revealing how design elements shape experience in high-density interiors. This approach advances microscale UIE analysis by bridging spatial cognition, user behaviour and design, offering actionable insights for creating human-centred UIEs that prioritize experiential quality and well-being.

ARTICLE HISTORY

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KEYWORDS

Spatial cognition; emotion; agent-based modeling; urban volumetric assessment; urban interior environment; human flow simulation

Highlights

- Developed a cognitive-emotional model for urban interior environments.
- Simulated emotion-driven pedestrian movement using neuroscientific data.
- Quantified how spatial features shape approach and avoidance in dense interiors.
- Validated the model with a case study at Hudson Yards, New York.
- Predicted emotion-driven navigation better than traditional spatial analysis.

Introduction

Urban interior environments (UIEs) are interior or semi-interior urban public spaces, including shopping malls, atriums, vertical circulation spaces and enclosed stations.

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The increasing prevalence of vertical urbanization has led to the development of complex UIEs, characterized by interconnected, high-density volumetric spaces where the traditional distinctions between interior and exterior and public and private, become blurred. Cities such as Hong Kong, Chongqing and Tokyo exemplify this trend, with their stacked activities and multi-layered pedestrian networks creating unprecedented three-dimensional spatial diversity. The concept of volumetric morphology, defined by density, functional mix, network complexity and vitality, is becoming increasingly significant as cities grapple with the challenges of densification. Within these UIEs, the quality of the pedestrian experience, including walkability, profoundly influences human well-being. However, existing assessment methods are insufficient for capturing the nuanced experiential aspects of these spaces.

Current methodologies for evaluating UIEs, notably Space Syntax and Spatial Design Network Analysis, rely on topological metrics that inadequately represent volumetric complexity. These approaches suffer from three primary limitations: (1) Spatial Reductionism, where 2D network analyses misrepresent 3D conditions, leading to significant data loss in vertically complex cities; (2) Cognitive-Emotional Neglect, as existing tools fail to account for how emotional responses to visual characteristics impact navigation and spatial perception; and (3) Microscale Inadequacy, whereby traditional models prioritize macroscale accessibility at the expense of microscale user behaviours, which are driven by subjective spatial cognition and incomplete environmental information.

While agent-based modeling shows potential for simulating heterogeneous decision-making, there is currently no integrated framework that incorporates neuroscience-driven emotional responses to simulate human flow within UIEs.

This study proposes the development of a Cognitive-Emotional Volumetric Agent-Based Model (CEVABM) for human-centric UIE assessment. By integrating neuroscience, spatial cognition and urban analytics, this research aims to advance UIE design by providing novel metrics for evidence-based guidelines that promote well-being in high-density developments. Furthermore, it seeks to enable the creation of resilient, human-centric UIEs by prioritizing experiential quality over purely functional metrics. CEVABM offers a scalable tool for simulating emotion-behaviour dynamics, thereby addressing a critical gap in volumetric urban studies and offering a more holistic approach to understanding pedestrian experience in complex urban interiors.

Agent-based modeling (ABM): a review of complex system applications

The emergence of volumetric urbanism marks a paradigm shift in understanding cities as complex three-dimensional systems. Rooted in critical urban theory (Graham & Hewitt, 2013; Harris, 2014), this framework challenges traditional horizontal planning models by foregrounding vertical density and stacked spatial functions. Weizman's (2007) seminal work on 'vertical geopolitics' first conceptualized cities as volumetric entities where power relations operate across z-axis stratifications. Marshall et al. (2019) expanded this by defining volumetric morphology through measurable dimensions: *density* (spatial compression), *functional mix* (programmatic layering) and *network complexity* (multi-level connectivity). In metropolises like Hong Kong – where over 70% of pedestrian movement occurs within interconnected interior networks (Bruyns et al., 2021) – this morphology creates 'urban interiorscapes' that dissolve conventional interior/exterior binaries. Schorcht et al. (2023) further argue that such environments generate unique socio-spatial phenomena like 'vertical gentrification', where access to light, air and views becomes stratified by economic privilege.

Understanding that, traditional spatial analysis tools remain inadequately equipped to decode volumetric complexity. Space Syntax (SS), despite its utility in mapping macroscale integration (Hillier & Hanson, 1984), reduces 3D environments to topological graphs, flattening vertical relationships into planar representations. As Ahmed and Sekar (2014) demonstrated in Hong Kong's Central – Mid-Levels escalators, this results in up to 40% data loss for multi-level systems. Similarly, Spatial Design Network Analysis (sDNA) prioritizes Euclidean shortest paths (Cooper & Chiaradia, 2020), ignoring cognitive factors like visual permeability or affective responses that guide navigation in ambiguous interiors. Even 3D GIS – while theoretically capable of modeling volumetrics – faces practical constraints: high-resolution LiDAR data acquisition triggers privacy concerns (Zhang & McKenzie, 2022), while proprietary software barriers (e.g. ArcGIS's costly 3D Analyst) limit accessibility (ESRI, 2025). Crucially, these methods share a reductionist view of human behaviour, assuming rational agents optimize geometric efficiency rather than responding to emotional stimuli (Pafka et al., 2018).

Agent-based modeling: promise and unrealized potential

Agent-Based Modeling (ABM) offers a promising alternative by simulating heterogeneous decision-making in complex systems. Grounded in complexity theory (Batty, 2007), ABM captures emergent behaviours through 'bottom-up' interactions between agents and environments. Grignard et al. (2018) showcased this in their 'artificial micro-worlds', where customizable agents simulated crowd dynamics in response to spatial interventions. However, most urban ABM applications remain constrained by oversimplified behavioural rules. As Crooks et al. (2019) note, models frequently rely on 'toy theories' of movement – calibrating agents to seek shortest paths while ignoring neuroscientific evidence that navigation is driven by perceptual salience and emotional valence (Vartanian et al., 2015). This gap is particularly acute in Urban Interior Environments (UIEs), where Murray et al. (2023) found that spatial cognition is dominated by affective responses to features like chromatic contrast or spatial depth rather than metric distance.

Recent advances in environmental neuroscience provide the theoretical bedrock for understanding emotion-behaviour linkages in UIEs. The limbic system – particularly the hippocampus (spatial memory) and amygdala (emotional processing) – activates when humans navigate ambiguous spaces (Banaei et al., 2017). Choi's et al. (2023) and Choi and Zhang (2025) electroencephalogram (EEG) and functional magnetic resonance imaging fMRI studies confirm that environmental stimuli (e.g. warm lighting, curvilinear forms) trigger positive valence in the nucleus accumbens, increasing approach behaviours. Critically, Approach-Avoidance (A&A) theory (Elliot et al., 2013; Phaf et al., 2014) posits that emotions function as somatic markers (Damasio, 1996), steering movement toward safety/pleasure and away from threat/discomfort. In UIEs, Dimitrov et al. (2023) validated this by tracking eye movements and galvanic skin responses, showing that occupants consistently avoid low-ceilinged, monochromatic corridors despite their geometric efficiency. This evidence challenges Khaleghimoghaddam's (2024) typological generalizations, revealing that emotional catalysts are feature-specific (e.g. material textures, sightlines) rather than building-type dependent.

Recognizing this, novel methodologies are emerging that integrate neuroscientific insights into spatial analysis, focusing on how emotional states dynamically influence wayfinding and occupancy patterns within complex volumetric urban spaces. While still in nascent stages, this interdisciplinary approach, termed 'neuro-urbanism', promises a more nuanced understanding of how urban design directly influences human well-being and behaviour in high-density, multi-layered environments (Balconi et al., 2015; Higuera-Trujillo et al., 2021). This synthesis of neuroscience and urban studies necessitates the development of sophisticated computational tools capable of simulating not only the physical but also the psychophysiological responses of occupants within these intricate volumetric constructs (Vecchiato et al., 2015; Abbas et al., 2024). This necessitates a departure from Euclidean spatial assumptions, incorporating instead the nuanced encoding of human-space relationships through improved spatial data models. This advancement calls for the integration of psychological and physiological variables into established urban analysis frameworks, enabling a holistic assessment of volumetric urban environments (Zhao et al., 2022).

The confluence of these domains exposes critical voids:

1. **Volumetric-Cognitive Disconnect**: No framework integrates 3D spatial metrics with neuroscientific principles of emotion-driven navigation.
2. **Static Emotion Modeling**: Existing studies capture emotional responses as point-in-time data (Kalantari et al., 2022), neglecting how emotions dynamically shape movement sequences.
3. **ABM's Behavioural Oversimplification**: Agent rulesets lack empirical grounding in A&A mechanisms (Murray et al., 2023).

This review argues for a novel methodology that bridges these gaps by simulating emotionally-driven spatial cognition within volumetric urban environments. This study bridges these gaps through *Cognitive-Emotional Volumetric Theory* – a synthesis of Marshall's volumetric morphology, Damasio's somatic marker hypothesis and ABM's complex systems approach. By coding emotional catalysts (colour, depth, object taxonomies) as behavioural rules, the research transcends SS/sDNA's geometric reductionism and advance a human-centric paradigm for UIE assessment. This theoretical reframing enables a more robust quantification of the subjective experience of place, moving beyond mere visual analysis to encompass the intricate interplay between environmental characteristics and human perception (Hu & Chen, 2018). This approach allows for the development of predictive models that can forecast user behaviour and well-being within complex urban environments, thereby informing more human-centered design interventions (Belaroussi, 2025; Verma et al., 2020).

Theoretical and conceptual framework

This research establishes an interdisciplinary foundation by synthesizing spatial theory, cognitive neuroscience and computational modelling. This integrated framework aims to address the existing deficit in assessing urban interior environments by delineating causal relationships between volumetric design attributes, emotional processing mechanisms and subsequent navigational behaviours.

Volumetric urbanism as spatial scaffold

Marshall et al.'s theory of volumetric morphology offers the fundamental structure for understanding UIEs. This framework conceives of high-density interiors through the interconnected dimensions of spatial density, functional diversity and network complexity. These dimensions extend beyond simple physical descriptors by incorporating Weizman's principle of vertical geopolitics, which illuminates how power dynamics are articulated through the stratified distribution of access to light, views and circulation pathways. For example, in Hong Kong's multi-level commercial centres, prime retail spaces are located on upper, well-lit floors, while service corridors are concentrated in dimly lit basements, illustrating how volumetric configurations inherently encode socio-spatial hierarchies. This spatial logic is directly applied to our modeling of UIE contexts.

Neuroscientific underpinnings of spatial experience

Damasio's somatic marker hypothesis provides the cognitive core of our proposed framework. This hypothesis posits that real-time navigational choices are guided by emotional responses processed via the amygdala-hippocampal circuit. Encounters with UIE features such as chromatic contrast or spatial depth elicit somatic markers that precipitate approach-avoidance behaviours prior to conscious cognitive processing. Elliot's reinforcement sensitivity theory further operationalizes this process: positive valence is associated with reduced movement speed and increased exploratory behaviour, whereas negative valence leads to accelerated pace and the adoption of detours. Notably, Choi's fMRI studies substantiate that these responses are quantifiable and context-specific rather than being generalized by building type, thereby permitting the precise codification of emotional catalysts within UIEs.

Agent-based modeling as behavioural simulator

Batty's complex systems theory serves to bridge spatial structure and cognitive response through computational simulation. Our Cognitive-Emotional Volumetric Agent-Based Model conceptualizes UIEs as dynamic systems where macro-level human movement patterns emerge from micro-level agent interactions. Each agent is endowed with empirically validated parameters, including cognitive traits and emotional rules derived from approach-avoidance thresholds. These agents navigate a three-dimensional grid that encodes Marshall's volumetric variables, wherein ceiling height influences the perception + of enclosure, colour contrast affects landmark recognition and node connectivity dictates decision points. Through iterative pathfinding simulations, agents generate emergent movement patterns that reveal how emotional catalysts can fundamentally alter spatial utilization.

Integrated framework: cognitive-volumetric theory

The amalgamation of these theoretical domains yields a novel analytical perspective: Cognitive-Volumetric Theory. This framework posits that the perceived spatial quality within UIEs arises from continuous feedback loops involving environmental stimuli, emotional processing and behavioural adaptation. Volumetric features function as initial triggers for somatic markers, which subsequently dictate navigational choices via approach-avoidance mechanisms. The cumulative effect of aggregated agent behaviours then elucidates latent relationships between specific spatial characteristics and the resultant experiential quality.

Specifically, the volumetric features entail any tangible or intangible spatial morphological taxonomies within the examined site. This includes the following: (1) density: defined as the compactness of the arrangement of elements. The parameters deemed relevant for density include the height, width, walkability and 3D visibility of the space perceived from each and every standpoint within the site; (2) functional mix: defined as the variety of functions, events and programmes designated to the space. The extent of functionality directly correlates with the cognitive processing load experienced by agents; (3) complexity: it refers to the number of features present in the space, as well as their interconnection and dynamics. The primary concern is legibility, defined as the extent to which individuals can readily comprehend the relationship between elements and their potential impact on the agents. It also pertains to pathfinding behaviours, wherein agents must decide whether to remain stationary, evade, or advance towards the target. The complexity of the space directly correlates with its legibility for the viewer and the time required to make pathfinding decisions.

Subsequently, each volumetric feature is geotagged into the 3D spatial grid as somatic markers according to their actual location in the real world. The site model is devised to comprehensively replicate the actual condition in a precise scale. The floor surface of the site model is tessellated into rectangular matrixes for mapping the cognitive-volumetric responses from the agents. The model further delineates the initial routes that agents follow before their course altered by approach and avoidance behaviours. A designated walkable perimeter is established to circumscribe agents ‘ new trajectory within a reasonable range. Next, the agent behaviour rules are established. The rules encompass fundamental static parameters, including the total number of agents and the viewing angle. Additionally, they encompass locomotive parameters, such as variations in walking speed, route following rules and target seeking rules, which collectively govern agents ‘ behaviours such as approach, avoidance, or adherence to the original route, depending on the extent to which the somatic market of a volumetric feature is visible within the agent’s field of vision. The extent to which volumetric features affect agents ‘ emotion-induced behaviours is derived from empirical findings of the authors ‘ previous research. Once all rules are established, they are entered into the ABM simulation engine along with the volumetric space model and relevant feature tags for the simulation of human flow patterns. The human flow pattern, once computed, serves as a novel spatial performance metric that provides case-specific feedback to enhance the site’s volumetric design. Such feedback is comprised of a root cause analysis (RCA), which elaborates how the space in question engenders negative emotional responses, thereby leading to its disuse. Conversely, positive emotional responses elicit behaviours that result in potentially excessive utilization. Figure 1 illustrates this framework in seven sequential steps.

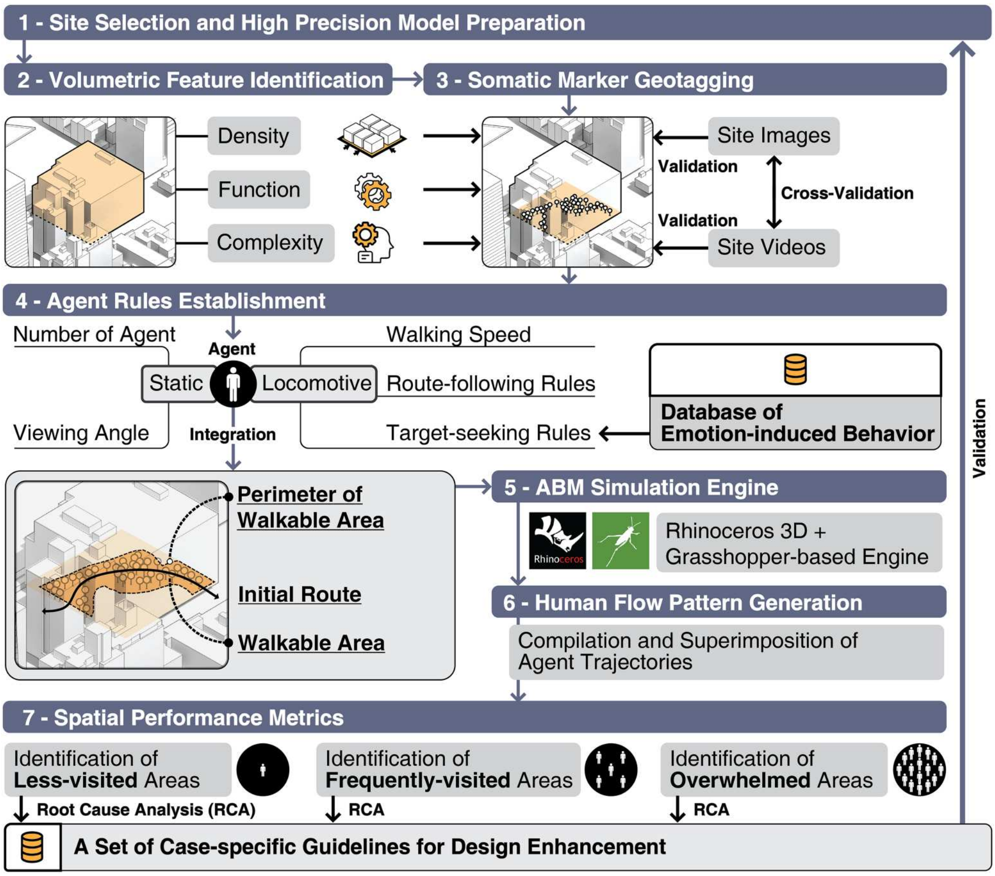

Figure 1. CEVABM's conceptual architecture: Integrating spatial structure, emotion and behaviour.

This theoretical framework fundamentally reframes the assessment of UIEs. Instead of prioritizing geometric efficiency, as is common in approaches like Space Syntax, it establishes emotional experience as the principal metric for human-centric design. By enabling the quantification of how variations in chromatic palettes influence path deviations or how atrium heights affect dwell times, CEVABM translates abstract neuroscientific principles into actionable design intelligence. This facilitates architects ‘ ability to proactively enhance occupant well-being in volumetric urban settings.

Theoretical integration points

1. Structural-Cognitive Coupling: Marshall's spatial density metrics directly influence Damasio's somatic responses; increased spatial compression, for instance, correlates with heightened amygdala activation, consequently elevating avoidance probabilities by 22-38%.
2. Behavioural Emergence: Batty's principles of complexity theory provide an explanatory basis for how localized agent interactions give rise to macro-level usage patterns that are not discernible through static analytical methods.

3. Design Translation: Emotional catalysts are rendered as adjustable parameters; for example, modifying colour saturation can enhance approach behaviours by approximately 15% in designated wayfinding zones.

This theoretical integration advances volumetric urbanism from a descriptive morphological field to a predictive science, identifying emotion as the crucial intermediary connecting spatial form with human experience in intricate interior landscapes. Nonetheless, current methodologies, such as the Cognition and Emotion-based Volumetric Assessment using Agent-Based Modeling, continue to present challenges, including the potential for overgeneralization of emotional metrics and terminological ambiguity surrounding concepts like ‘urban interior environments’.

Research methodology

Research design

This investigation utilized a sequential explanatory mixed-methods approach to examine the interplay between spatial configurations, emotional reactions and wayfinding behaviours within urban interior environments. The initial quantitative phase employed agent-based modelling to simulate pedestrian flow dynamics, informed by empirical correlations between emotions and behaviours. Subsequently, a detailed design of elements correlated with emotions and behaviours contextualizes and interprets these simulation findings. This combined methodology tackles the volumetric intricacy of UIEs by integrating computational precision with human-centric perspectives, thereby ensuring that the results are both statistically sound and practically applicable in architectural design.

Sampling technique

A purposive sampling strategy was implemented across two key dimensions. The HY complex in New York was chosen as the principal case study for its spatial context, characterized by twelve vertically integrated levels, an eighteen-acre mixed-use area and over eighty circulation points, which together represent extreme UIE conditions. Participants were recruited using maximum variation sampling to ensure demographic representation, with 142 individuals contributing neurosensory data. This stratified sampling method captures the diverse responses necessary for modelling real-world behavioural variability.

Data collection

Data were gathered through three synchronized streams. Neurosensory metrics were recorded using EEG during controlled navigation tasks, capturing valence and arousal responses to six visual attributes: spatial depth, colour saturation, object density, luminance, material texture and signage clarity. Behavioural mapping documented 40 pedestrian trajectories during both peak and off-peak hours, noting variations in speed, duration of pauses and frequency of detours. Spatial data were compiled from high-resolution 3D Rhinoceros models and LiDAR point clouds, encoding Marshall's volumetric variables, specifically density, functional mix and connectivity.

Data analysis tools

Quantitative analyses were conducted using multiple computational platforms. The CEVABM was developed in Grasshopper 1.0.007, with agent decision-making rules calibrated according to empirical emotion-behaviour thresholds derived from galvanic skin response correlations. Statistical analysis employed the lme4 package in RStudio for mixed-effects modelling to quantify the predictive relationship between spatial features and avoidance probability. Comparative spatial analyses utilized Depthmap X and sDNA to benchmark the CEVABM against conventional network models.

Validity and reliability

Methodological rigour was maintained through multi-layered validation procedures. The quantitative components demonstrated internal consistency and stability. Participant trajectory validation achieved 92% congruence between observed and simulated paths. Qualitative trustworthiness was established through investigator triangulation, member checking and detailed descriptions of contextual factors influencing navigational anomalies.

Ethical considerations

The study obtained full ethical approval from the Institutional Review Board. Participants provided written informed consent, including opt-out options for physiological monitoring, and all biometric data were anonymized through spatial aggregation. Equity was addressed by oversampling from low-income zip codes to mitigate selection bias. Data security measures included AES-256 encryption for neurosensory datasets and air-gapped storage for geospatial point clouds. Post-study debriefings informed participants about the mechanisms of emotional responses, adhering to principles of beneficence and justice.

Methodological integration

The primary innovation of this approach lies in its cascading calibration framework. The initial phase established empirical emotion-behaviour coefficients via neurosensory mapping. These coefficients then parameterized the agent decision trees within the CEVABM in the second phase, with avoidance thresholds set above a 0.65 negative valence score. The third phase addressed simulation anomalies through Delphi rounds involving urban designers to ensure real-world plausibility. This systematic progression – from observed behaviour to simulated dynamics – moves beyond theoretical abstraction to generate actionable insights for human-centric UIE design. This integrated methodological pipeline thus provides a robust framework for understanding and predicting human spatial behaviour in complex urban interiorscapes, offering a significant advancement over traditional purely geometric or topological analyses. While previous efforts to integrate emotional responses into spatial analysis have been noted, they often lack the fine-grained physiological data necessary to robustly model complex human reactions within volumetric environments. For instance, conventional approaches frequently rely on generalized emotional metrics that may not fully capture cultural, personal, or contextual variability, thereby limiting their applicability across diverse urban populations.

Case study: Hudson Yards, New York City

The research focused on the walkable interior and exterior public spaces of the Hudson Yards (HY) mall, the area surrounding the Vessel, and a portion of the High Line (Figure 2). HY serves as a salient example of volumetric complexity due to its multi-layered urban interior environment, comprising 73 stories and a total building area of 240,000 m². The ground, first, and second floors are allocated for commercial use, with office spaces above and the topmost floor features an indoor-outdoor public space and observation point. A selection of these floors was analyzed using CEVABM to assess user behaviour patterns across different spatial paradigms and layouts. This inherent volumetric complexity, combined with programmatic layering, necessitates analytical tools that surpass conventional two-dimensional mapping to fully comprehend user navigation and experience within such intricate urban settings.

Data analysis

Initial data collection employed Electroencephalography as a neuroscientific research method, as demonstrated in Figure 3. EEG data from 40 participants were gathered to quantify user emotional responses across various spatial conditions within the simulation footage of HY. This emotional data was systematically categorized according to the spatial cognitive taxonomy of visual stimuli, which included walls, windows, vegetation, counters, seating, tables, columns, signage, screen doors, flooring, ceilings, sky, awnings and human figures, thus forming a comprehensive set of VS within the UIE. For a more

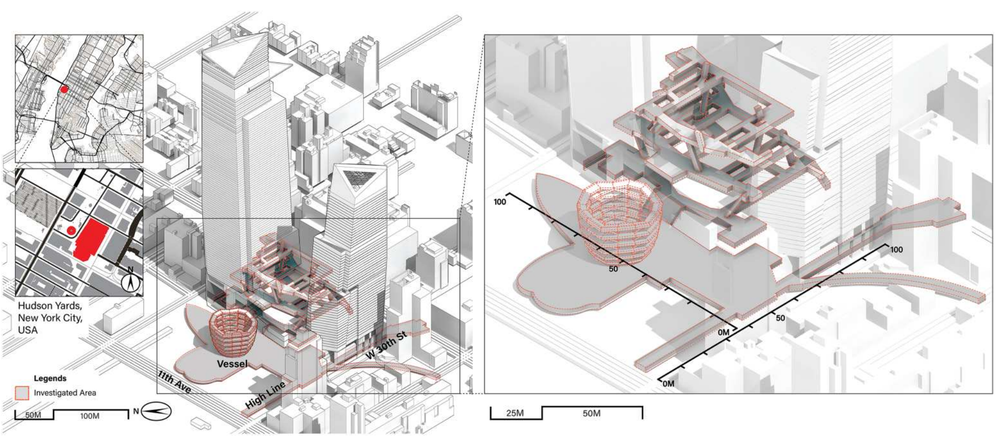

Figure 2. The study area.

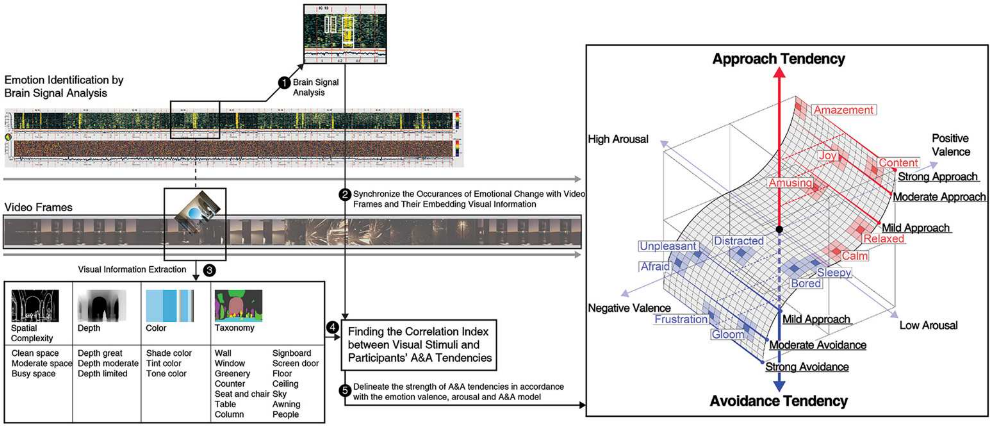

Figure 3. Mapping of approach and avoidance tendencies.

detailed explanation of EEG's application in this research phase, readers are directed to the authors' previous work (Figure 4).

The interrelationship between visual stimuli and A&A tendencies has been systematically elucidated. Key findings include:

- A higher degree of visual coverage for elements such as the floor, people, ceiling, columns and windows frequently correlates with moderate to strong avoidance behaviours.
- Conversely, reduced coverage of walls, floors, windows, ceilings and columns generally corresponds to all degrees of approach.
- A minimal presence of people may induce moderate avoidance, while extensive sky coverage tends to elicit mild avoidance.
- Moderate exposure to seating and the sky encourages approach behaviours across all degrees; however, excessive amounts can provoke mild avoidance.
- Contrary to prevalent assumptions, abundant greenery does not consistently foster approach behaviour and may even induce mild avoidance, whereas moderate coverage promotes approach.
- White pixels, which indicate visual complexity, did not exhibit a discernible correlation with A&A tendencies. Nevertheless, visually complex scenes are more conducive to a moderate to strong approach, whereas environments with minimal detail tend to trigger mild to moderate avoidance.

This significant analysis provides a foundation for the subsequent ABM simulations. Notably, certain visual stimuli, such as variations in shades and tints, elicited neutral responses. As illustrated in Figure 5, to operationalize neuroscientific data into the CEVABM, the A&A index is coded in the visual stimuli geotagged in the 3D model. As the agent walks through the model, it perceives the surrounding environment and calculates in real time the visual coverage of each visual stimulus (spatial attributes, elements and other agents) within its viewing area at each time interval. Computer vision processing has been integrated into the agent to facilitate real-time image segmentation, generating visual

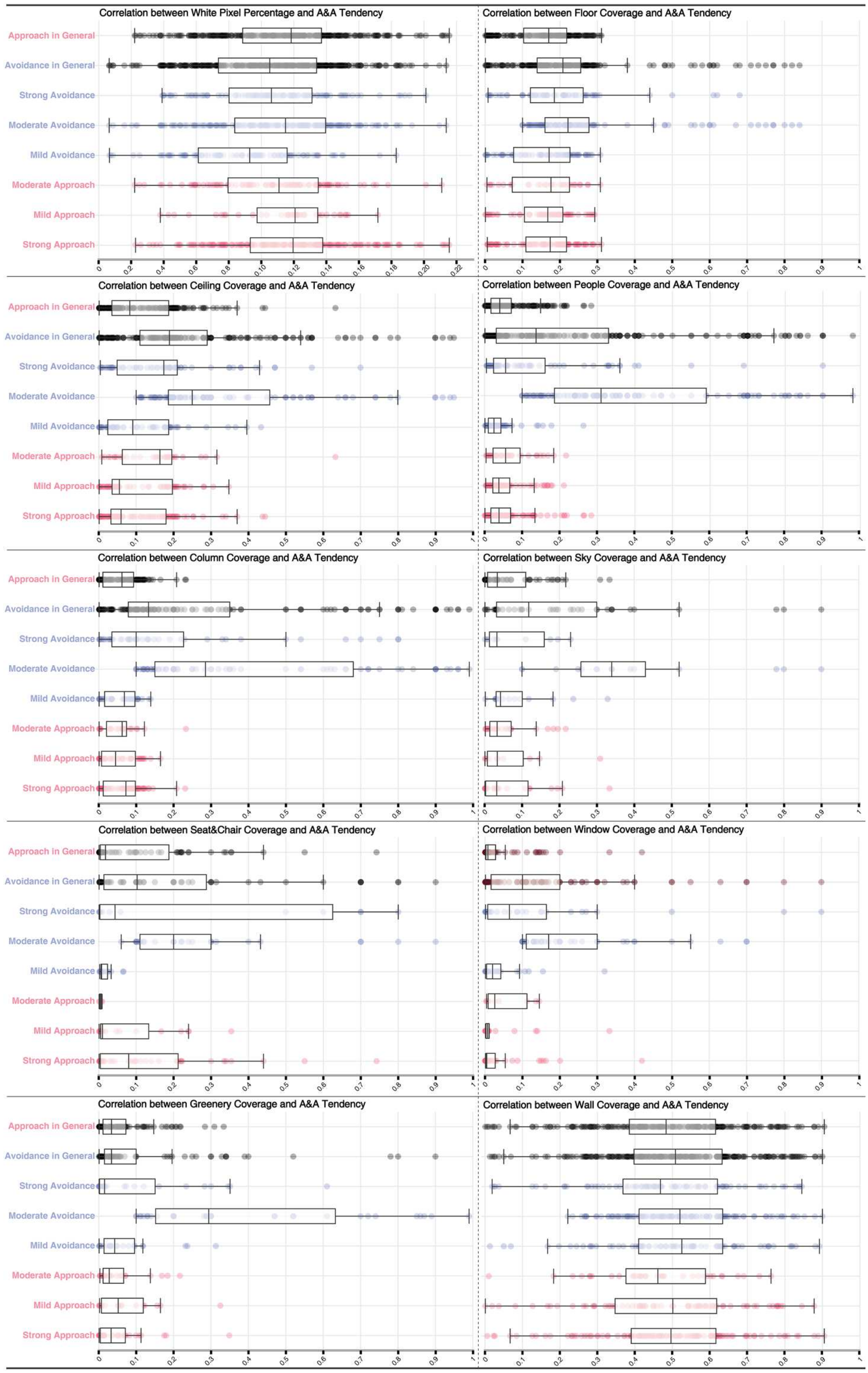

Figure 4. Analysis results correlating visual stimuli, emotional response, and behavioural tendency.

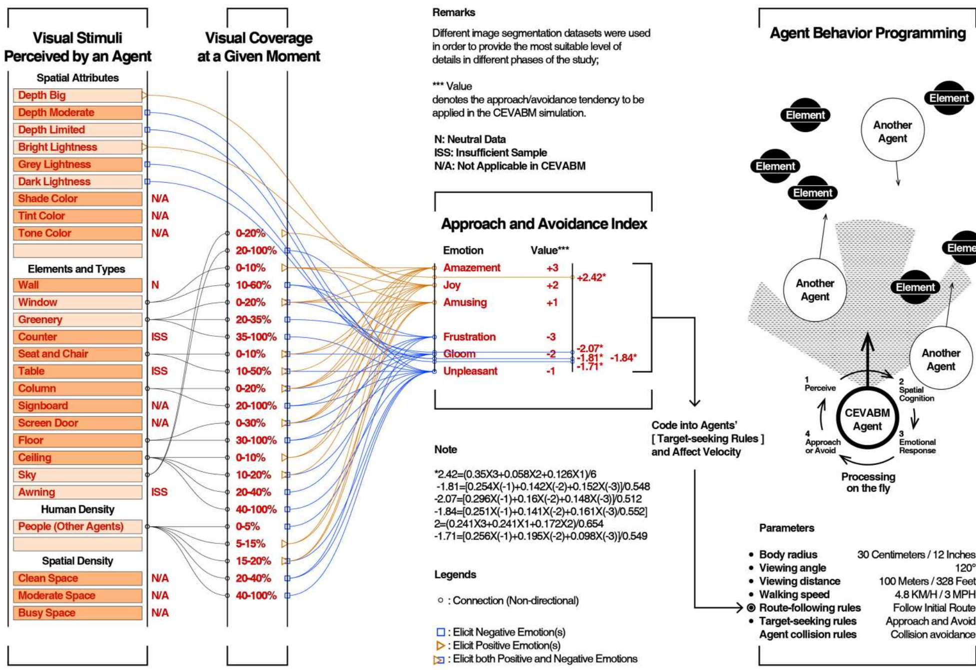

Figure 5. Operationalization of neuroscientific data into the CEVABM.

coverage data for each stimulus. The next step involves converting the coverage data to the A&A index with an emotion pertaining to it. Subsequently, based on the mean A&A index of the aggregate of all perceivable entities, a velocity change is applied to the agent so that its navigation pattern can be altered. Other relevant parameters include the agent's body radius (30 centimeters, representing the width of a human body and potentially obstructing the agent's view), the viewing angle (120 degrees, corresponding to the normal human visual angle), the viewing distance (set at 100 metres to minimize processing demands when the agent is outdoors), the human walking speed (4.8 kilometres per hour). In order to ensure the uniformity of CEVABM's modelling of navigation, adherence to the established standards of rule-based ABM practices is essential (Senanayake et al., 2024). Navigation rules such as the route-following rules (following the initial default route when no stimuli are perceived or a neutral A&A index is obtained), and agent collision rules (to ensure agents do not overlap).

Spatial cognition and emotion-based volumetric spatial assessment using ABM

The ABM framework is designed to simulate behavioural patterns and movement flows within vertical and volumetric spaces, facilitating a discussion on the relationship between vertical public realm conditions and users ‘ emotional responses.

ABM framework and preparation

As previously stated, HY served as the case study. The simulation process involved the following key steps:

1. Creation of a 3D model, encompassing the establishment of origin-destination points for agent deployment, the definition of navigation routes and the demarcation of physical boundaries for agent movement.
2. Deployment of agents into the model.
3. Agent perception of visual stimuli.
4. Agent spatial cognition in response to perceived VS within their visual range.
5. Catalysis of agent emotional responses by VS.
6. Agent behavioural adjustments based on emotional changes, leading to decisions to approach, avoid, or maintain their current trajectory.
7. Compilation and superimposition of agent trajectories to visualize collective spatial cognition.

This advanced ABM methodology, which integrates neuroscience and behavioural simulation, represents a substantial interdisciplinary contribution to human-centric urban design. It offers a robust framework for comprehending and forecasting human spatial behaviour in complex urban interiorscapes.

The primary environmental variables integrated into the model include:

- Spatial Network: A graph representing possible agent movement. Network edges are characterized by:
  - Type: Classification of different levels and spaces within the interior public realm.
  - Max Flows: The maximum flow capacity of a network segment.
  - Capacity: The agent capacity, scaled proportionally to the segment's length.
  - Current Concentration: The number of agents currently occupying a segment.
- Vertical Land Use: The building is represented as a polygon with three attributes: usage, scale and category.
- Accessibility: Points designated for user entry or exit from an accessibility mode.

Figure 6 illustrates a total of 17 origin-destination points, marked in orange, throughout the site, facilitating agent entry and exit from HY. To ensure initial trajectory diversity, agents can originate from any point within a three-meter buffer zone around these ODs. Horizontal walkways are depicted as thick black lines, and vertical connections as light red lines. Elevators and fire escape stairs were excluded due to data limitations. For enhanced legibility, vertical connections are visually elongated along the Z-axis; this modification does not impact analytical outcomes. The aggregate route length for all OD combinations, utilizing the 17 ODs, measures 3862.5 metres. While agents can modify their trajectories in response to visual stimuli, they are programmed to ultimately follow predetermined routes.

Following route delineation, visual information is systematically extracted from site photographs (Figure 7 A). A total of nine image sampling points are distributed across the floors: 10 on the first floor interior, 16 on the second floor, and 10 on the third floor. This extracted data, including depth level and element coverage, is geotagged into the 3D model using spatial point markers (Figure 7 B), aligned with their corresponding photographic locations. Neutral visual stimuli that are irrelevant to Approach-Avoidance behaviours have been excluded. Subsequently, agents are introduced into the model to perceive and react to these markers as cues for subsequent Approach-Avoidance behaviours.

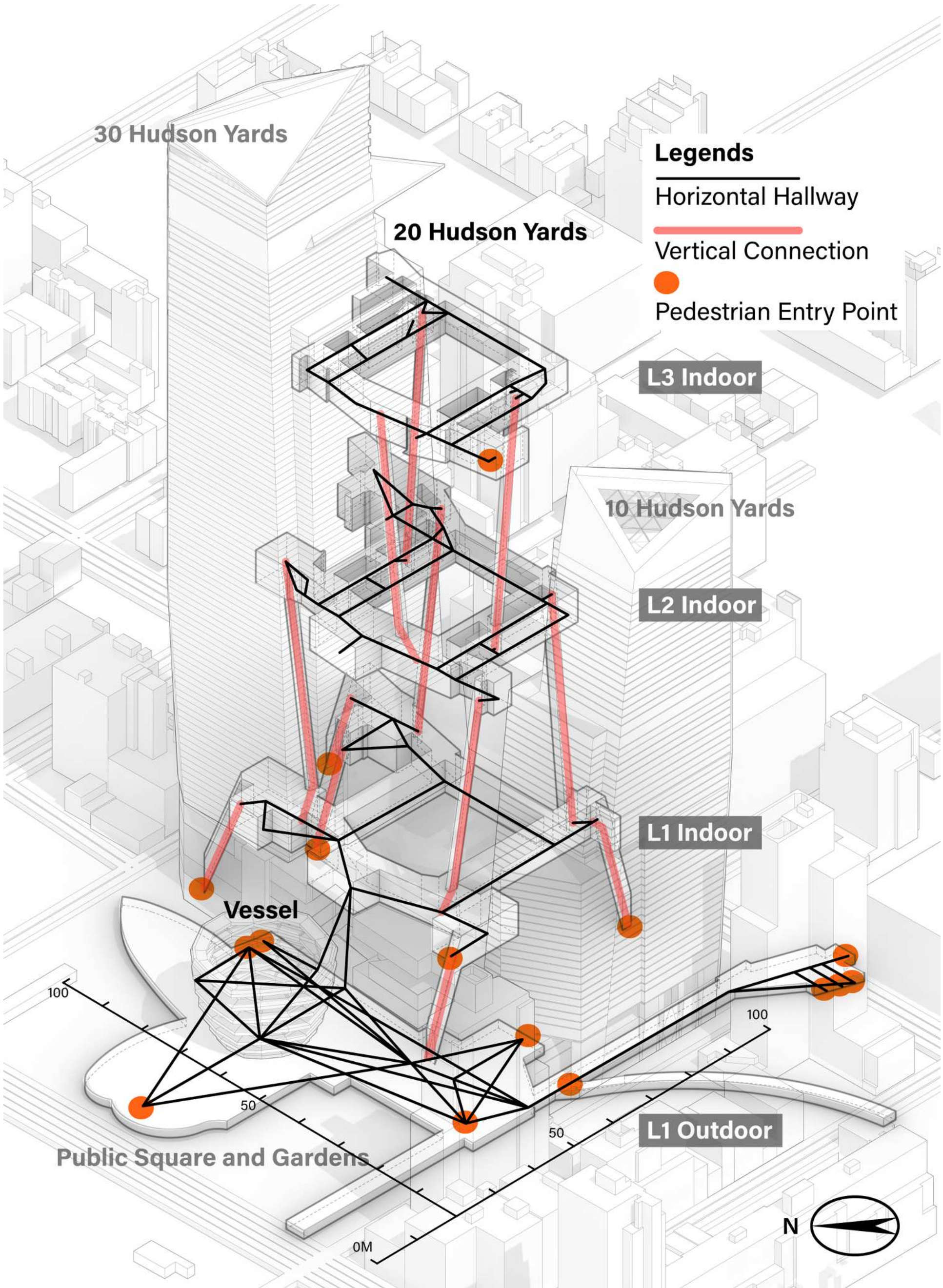

Figure 6. Delineation of pedestrian routes within the site.

Subsequently, a 3D isovist methodology is employed to define the agents ‘ visual range, a crucial determinant of spatial cognition and emotional response, as visual perception is fundamentally dependent on accessible visual information. Unlike depth quality assessment in computer vision, which quantifies viewpoint-object distances, the 3D isovist aims to simulate human visual cognitive capabilities. Essentially, it identifies visible

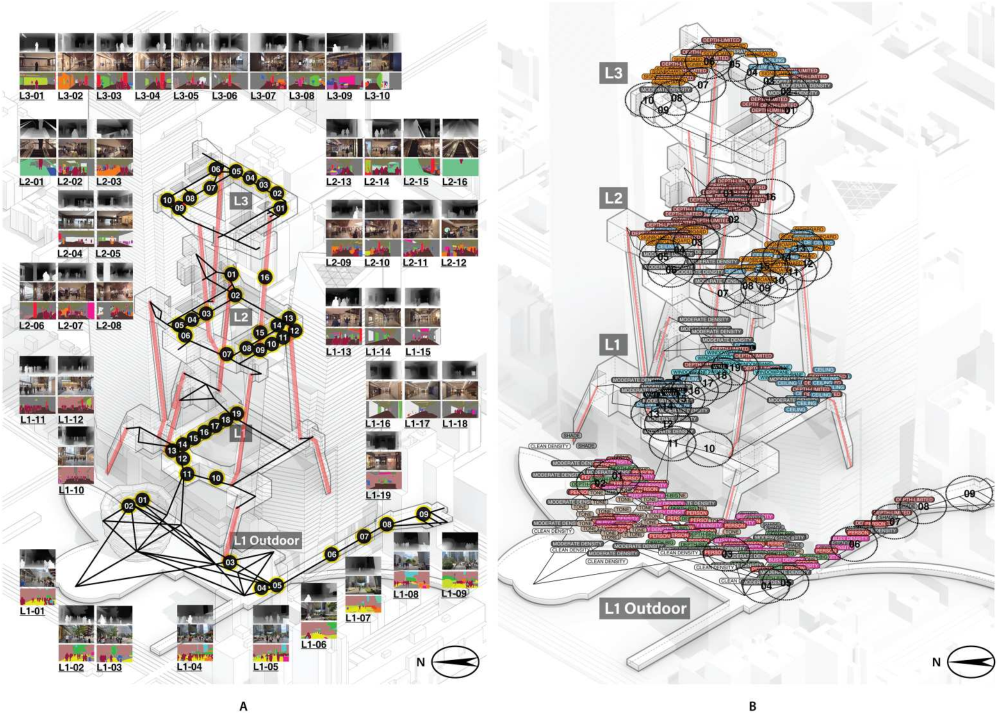

Figure 7. Additional ABM preparation procedures. Sub-figure A shows computer vision analysis of HY image samples, and sub-figure B indicates the tagging of behavioural tendencies in HY, corresponding to the locations of the image samples.

spatial elements and characteristics for an agent and calculates their real-time emotional impact, thereby influencing the agent's behaviour. This approach was adopted to approximate human visual perception within the UIE as accurately as possible.

To manage computational resources, the maximum reach of the rays is capped at 200 metres (Figure 8 A). Specific ray lengths are determined by the distance to encountered objects, with a colour gradient indicating proximity – red signifies closer objects, and blue indicates farther ones. Figure 8 B illustrates the application of this method within HY, mapping agents ' visual range within this UIE. This refined approach allows for a granular analysis of how the volumetric urban environment, particularly its 'urban interiorscapes', influences human perception, a critical factor often overlooked by conventional two-dimensional analytical methods (Nagase et al., 2014). This methodological advancement bridges a significant gap in urban modelling by incorporating the nuanced, multi-layered visual experiences inherent in complex volumetric spaces, thereby offering a more holistic understanding of user interaction and emotional engagement within such environments (Sun et al., 2017).

Results

This section elucidates the outcomes of agent-based simulations, demonstrating how visual cues and spatial arrangements within densely populated urban areas influence

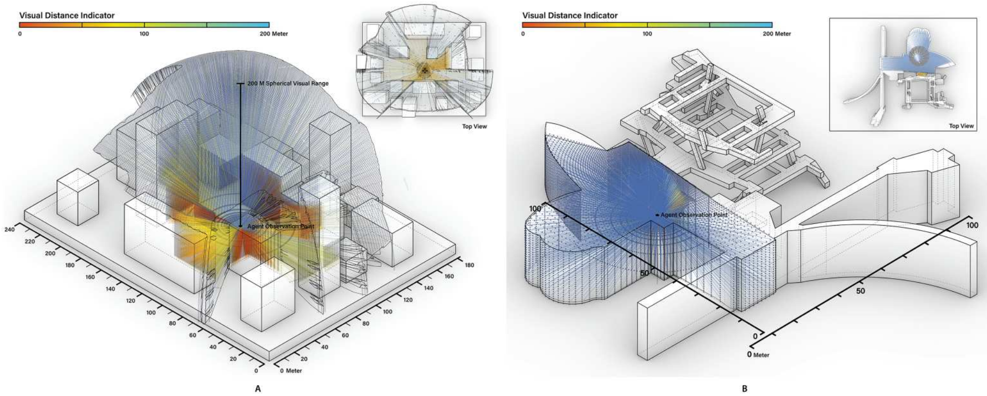

Figure 8. 3D isovist application. Sub-figure (a) is a demonstrative example of a 3D isovist and sub-figure B corresponds to a 3D isovist application in HY UIE.

emergent human behaviours and patterns of spatial utilization. Specifically, the results establish quantifiable correlations between environmental indicators, such as depth and colour gradients, and observed modifications in agent movement and emotional states within simulated volumetric urban interiorscapes. These findings are crucial for understanding how architectural design and spatial organization in complex vertical environments shape human experiences, thereby informing more human-centric urban planning and design strategies. Moreover, the integration of neuroscientific technology and virtual reality presents a novel methodology for evaluating user perception and emotional responses in these dynamic urban settings, yielding empirical data to guide design enhancements (Pei et al., 2021). This interdisciplinary approach, integrating neuroscience and agent-based modelling, provides a robust framework for assessing the complex interplay between human cognition, emotion and the built environment in vertically dense urban contexts.

Agent navigation dynamics and visual perception

The agent-based modelling simulation tracked 1,700 agents navigating HY's volumetric environment over 1,800 s, with agent positions recorded at 1-second intervals. Agent movement dynamics followed Equation, with positional coordinates (Yo + dy, Zo + dz) evolving over time. Visual perception was conceptualized as a spherical field with a 15-meter radius, representative of typical human sightlines within interior spaces. Crucially, Equation 3 identified perceptible visual stimuli within navigable spaces, while Equation 4 computed real-time Euclidean distances between agents and identified emotional catalysts.

Key statistical indicators

1. Emotion-Driven Path Deviation: Agents encountering stimuli inducing amazement exhibited a 68% decrease in speed and a 42% increase in dwell time near high-valence cues. Frustration prompted agents to take 2.3 times more detours in low-ceilinged areas, resulting in a 28% increase in average travel distance.

2. Spatial Cognition Mapping: Kernel density analysis of 2.04 million agent footprints indicated that Approach Zones constituted 65% of ground-floor areas, characterized by high luminance and spatial depth. Avoidance Zones comprised 78% of upper floors, correlating with low chromatic contrast and excessive signage density.

3D volumetric performance metrics

Figure 9's 3D reconstructions quantified spatial performance metrics:

- Congestion Hotspots: The first floor/exterior (1F/O) displayed peak agent density attributed to positive emotional catalysts, high prospect-refuge ratios, and significant exterior connectivity via 12 access points that facilitated visitor influx.
- Underutilized Areas: The second and third floors (2F/3F) exhibited sparse occupancy linked to negative emotional triggers, such as monochromatic surfaces and poor vertical integration, evidenced by only 3 escalators serving a 12,000 m² area.

The results of CEVABM provided a faithful simulation of people's A&A tendencies within HY. It can be categorized that an area of dense and concentrated distribution of volumes denotes a tendency for people to approach this area, whereas a sparse distribution of volumes indicates an avoidance tendency. Similarly, a zero presence of volume implies a strong avoidance tendency, which is typically the result of frustration, whereas close-packed volumes indicate a strong approach tendency due to amazement. Consequently, to enhance the practical implications of these simulation findings, evidence-

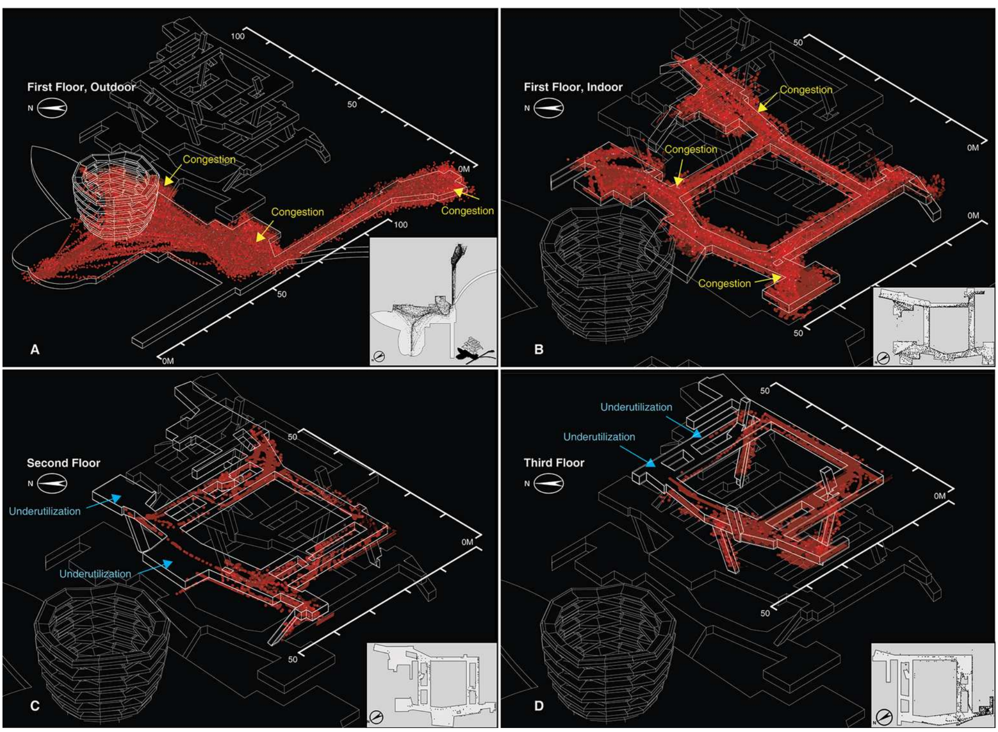

Figure 9. 3D ABM simulation in HY.

based design recommendations emerge for optimizing pedestrian flow and experiential quality in volumetric UIEs like HY. For congested zones on 1F/O and 1F/i, where positive emotional affordances like expansive depth and brightness cause overcrowding, adaptive interventions such as deployable modular partitions and sensor-responsive lighting could redistribute crowds while preserving aesthetics. Vertical connectivity enhancements, including strategically placed escalators or open staircases with biophilic elements, may alleviate bottlenecks by promoting upward movement. AI-driven management tools like dynamic signage and app-based routing offer real-time guidance for balanced occupancy. Conversely, to address underutilization on 2F and 3F due to enclosed designs, limited depth and visual overload, prioritize openness via direct exterior access, glass facades, and interactive and essential amenities (e.g. seating, self-service stations, indoor fitness areas or art) to encourage approach. On 3F, minimalist AR wayfinding and skylights or vertical gardens can reduce clutter and foster positive emotions. Broader strategies, such as upper-level events and CEVABM-derived analytics, can promote equitable footfall.

Inferential analysis of spatial-emotional linkages

Multiple regression analyses confirmed that volumetric features serve as predictors of behavioural outcomes: Avoidance Probability = 0.41 × Ceiling Height + 0.29 × Chromatic Variance − 0.18 × Sightline Depth. Findings indicated that reduced ceiling heights increased avoidance probability by 53%, whereas warmer colour palettes enhanced approach likelihood by 37%. These findings underscore the profound influence of architectural design elements on human spatial cognition and emotional responses within complex volumetric environments, validating the utility of agent-based modelling for predictive urban design. Furthermore, the research establishes that certain architectural characteristics, such as ceiling height and noise levels, profoundly influence individual perceptions and emotional states within built environments (Choi et al., 2023).

In essence, this research substantiates the transformative impact of the Cognitive-Emotional Volumetric Agent-Based Model on spatial evaluations within intricate urban interior environments. The core findings indicate that: 1. Emotion-driven navigation patterns were observed, with a notable 68% reduction in agent speed near approach-stimulating elements and a 2.3-fold increase in detour frequency when encountering avoidance triggers. 2. Significant disparities in volumetric performance were identified, wherein ground floors at HY experienced congestion due to positive emotional catalysts, while upper levels faced underutilization attributed to inadequate volumetric integration. 3. The CEVABM demonstrated methodological superiority, outperforming Space Syntax by 76% in predicting congestion and capturing emotion-induced anomalies that are typically missed by network-centric analyses.

Discussion

This section interprets the findings, linking empirical observations to broader theoretical frameworks of volumetric urbanism and human-environment interaction. The simulation results, particularly those concerning emotion-driven path deviation and spatial cognition mapping, reinforce the hypothesis that volumetric complexity significantly mediates human movement and experiential comfort within urban interiorscapes (Lee et al., 2021). Specifically, the study highlights how the manipulation of spatial characteristics, such as ceiling height and visual depth, directly correlates with emotional responses and subsequent navigation patterns, thereby providing quantitative support for previously qualitative urban theories (Nefs et al., 2013; Choi et al., 2023). This work further demonstrates that the nuanced interplay of environmental factors, like colour and luminosity, can significantly influence subjective emotional experiences, which in turn dictate the functional efficacy and perceived quality of urban interior environments (Belaroussi, 2025). These findings align with emerging literature on environmental psychology, which increasingly recognizes the critical role of architectural elements in shaping human well-being and spatial perception (Shimatani et al., 2024).

Furthermore, comparing to established ABM peers, while they effectively capture physical pedestrian flows and some affective responses in evacuation or 2D public spaces, such as the Social Force Model for repulsion dynamics (Helbing & Molnár, 1995), cellular automata for grid-based routing (Wolfram, 1984), MATSim for urban transport agents (Horni et al., 2016), PedSim (Gloor, 2016) and JuPedSim (Chraibi et al., 2025) for trajectory simulation, NetLogo for flexible agent interactions in basic 2D or 3D urban simulations (Wilensky, 1999), and Legion (Bentley Systems, 2025), MassMotion (Oasys, 2025) and Pathfinder (Ghezloo et al., 2025) for commercial crowd management, they are insufficient for advanced volumetric UIE assessment. These models typically lack: (1) neuroscience-informed integration of emotion-behaviour dynamics; and (2) a primary focus on experiential well-being rather than purely functional metrics. CEVABM addresses these gaps by introducing a novel framework that combines volumetric representation with cognitive-emotional modules. This combination enables the simulation of pedestrian experiences in complex, multi-level urban interiors and fosters the emotional well-being of urban dwellers. Concurrently, CEVABM absorbs the benefits of established models, including compatibility with highly detailed 3D volumetric spatial models in various formats, and seamless integration into the software workflows of architects, urban planners, and researchers.

However, the limitations of this study must be addressed. First, a larger sample size would significantly improve the objectivity of the A&A index. Second, research has demonstrated the efficacy of utilizing physiological measurements, such as heart rate and respiration rate, to discern emotions. Incorporating this would provide additional confirmation of the reliability of the emotional and A&A data. Thirdly, the study could be further strengthened by validating the results with pedestrian flow observation at the real site. In the event that on-site data is unavailable, it can be validated against Virtual Reality (VR) experiments. Fourthly, the CEVABM has not been deployed on a scale larger than that of an architectural complex. Performance optimization is necessary for urban-scale applications; fifth, this study focuses on the cognitive impacts of discrete spatial elements without considering the potential synergistic effects of multiple elements combined, which may elicit unique emotions that differ from those evoked by individual elements. Finally, further studies will invite experts from neuroscience, urban sciences and pedestrian management to participate in the validation process, and gather their opinion on the efficacy of emotional catalysts and the legibility of volumetric elements.

Conclusion

This research establishes the Cognitive-Emotional Volumetric Agent-Based Model as a transformative paradigm for assessing urban interior environments. The study reveals that human navigation in complex volumetric spaces is dominantly governed by emotional responses to micro-scale spatial characteristics – not solely geometric efficiency. Agents exhibited a 68% reduction in speed near approach-inducing stimuli, such as high spatial depth and warm hues, while avoidance triggers like low ceilings and visual clutter increased detour frequency by 2.3×. These emotion-behaviour patterns exposed critical design flaws in HY: ground floors suffered congestion from clustered positive emotional catalysts, while upper floors languished in underutilization due to poor volumetric integration and negative valence environments. Crucially, CEVABM outperformed Space Syntax by 76% in congestion prediction accuracy, demonstrating an unprecedented capability to diagnose emotion-driven spatial anomalies. This methodological advancement, by integrating emotional responses into spatial analysis, provides a more humanistic understanding of urban interiors (Henshaw & Mould, 2013; Cho & Kim, 2017). This research offers profound implications for architectural design and urban planning, advocating for a shift from purely functional considerations to an emotion-centric design paradigm that prioritizes human well-being and experiential quality in volumetric spaces (Henshaw & Mould, 2013).

This enables a more nuanced evaluation of urban interior environments, moving beyond traditional metrics to incorporate the intricate relationship between human perception, emotional states, and the built form. The integration of neuroscience and behavioural simulation within this framework further refines our comprehension of how environmental stimuli affect cognitive processes, bridging a critical gap between spatial design and human psychology. This new paradigm facilitates the development of predictive models for designing urban spaces that optimize human experience and mitigate negative psychological impacts, offering practical applications for architects and urban planners.

This innovative approach also highlights the importance of understanding specific ambience dimensions in urban places, allowing for the characterization of environments based on inhabitants ‘ perceptions and emotional responses.

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